

Can You Taco Tongue and Evil Eyebrow? Day 2



Are the events "Yes Taco Tongue" and "Yes Evil Eyebrow" independent?

1. Find class data from the previous lesson and fill in the following two-way table.

	Yes Evil Eyebrow	No Evil Eyebrow	Total
Yes Taco Tongue	15	5	20
No Taco Tongue	3	7	10
Total	18	12	30

2. Suppose we randomly choose a student from class.

Conditional probabilities

a. Given that the person selected is a Yes Evil Eyebrow, what is the probability that they are a Yes Taco Tongue? Write as a fraction, a decimal, and a percent.

Both → 15
Yes E.E. → 18
$= 0.833 = 83.3\%$ ← $P(\text{Yes T.T.} \mid \text{Yes E.E.})$
"given"
condition

b. Given that the person selected is a No Evil Eyebrow, what is the probability that they are a Yes Taco Tongue? Write as a fraction, a decimal, and a percent.

$\frac{5}{12} = 0.417 = 41.7\%$ ← $P(\text{Yes T.T.} \mid \text{No E.E.})$

c. Based on your two previous answers, what conclusion can you make?

People who can Evil Eyebrow are more likely to be able to Taco Tongue than those who can't Evil Eyebrow.

Definition: Two events are **independent** if knowing whether or not one event has occurred does not change the probability that the other event will occur.

3. Are the events "Yes Taco Tongue" and "Yes Evil Eyebrow" independent? Explain.



No. Knowing whether or not a person can Evil Eyebrow does change the probability that they can Taco Tongue.

4. Consider the data for all Seniors at EKHS.

	Yes Evil Eyebrow	No Evil Eyebrow	Total
Yes Taco Tongue	180	300	480
No Taco Tongue	20	100	120
Total	200	400	600

Find each of the following using the values in the table. Write as a fraction, a decimal, and a percent.

a. P(Yes Taco Tongue) = $\frac{480}{600} = 0.80 = 80\%$.

b. P(Yes Taco Tongue | Yes Evil Eyebrow) = $\frac{180}{200} = 0.9 = 90\%$.

c. P(Yes Taco Tongue | No Evil Eyebrow) = $\frac{300}{400} = 0.75 = 75\%$.

d. Are "Yes Taco Tongue" and "Yes Evil Eyebrow" independent? No

5. Suppose that the ability to Taco Tongue and Evil Eyebrow are actually **independent** events. Determine the following probabilities.

$P(A) = P(A|B) = P(A|B^c)$

P(A) a. P(Yes Taco Tongue) = 0.80 = 80%.

P(A|B) b. P(Yes Taco Tongue | Yes Evil Eyebrow) = 0.80 = 80%.

P(A|B^c) c. P(Yes Taco Tongue | No Evil Eyebrow) = 0.80 = 80%.

6. Fill in the table based on these probabilities.

80% of 200 → 160; 80% of 400 → 320

	Yes Evil Eyebrow	No Evil Eyebrow	Total
Yes Taco Tongue	160	320	480 ← 80% are Yes T.T.
No Taco Tongue	40	80	120
Total	200	400	600

7. How can you tell from a two-way table if two events are independent?

Check if the conditional probability of one event stays the same given whether or not the other event occurs.

Lesson 3.5 – Conditional Probability and Independence

QuickNotes

LT#1 Conditional Probability

The probability of one event given another event has occurred

$$P(A|B) = \frac{P(A \text{ and } B) \leftarrow \text{both}}{P(B) \leftarrow \text{given condition}}$$

"given"

LT#2 Independent Events

Knowing whether or not one event occurs does not change the probability of the other event.

If $P(A) = P(A|B) = P(A|B^c)$ then A and B are independent.

Check Your Understanding

The Pew Research center randomly selected 100 mothers in 1976, 1994, and 2014 and asked them how many children they have. The two-way table summarizes the responses.

Number of children	Year: 1976	1994	2014	Total
1 child	11	21	22	54
2 children	24	43	41	108
3 children	25	23	24	72
4+ children	40	13	13	66
Total	100	100	100	300

Suppose we randomly select one of the survey respondents. Define events C: have 4 or more children, S: 1976, N: 1994, and F: 2014.

1. Find $P(C \mid S)$. Interpret this value in context.

$P(C|S) = \frac{40}{100} = 0.40$ Given that the mother was surveyed in 1976, there is a 0.40 probability she had 4+ children.

2. Given that the chosen mother was not surveyed in 1976, what's the probability that she has 4 or more children? Write your answer as a probability statement using correct symbols for the events.

$P(C|S^c) = \frac{13+13}{100+100} = \frac{26}{200} = 0.13$

3. Are the events "Surveyed in 1976" and "4 or more children" independent? Explain.

No. Mothers surveyed in 1976 are more likely to have 4+ children than mothers surveyed in not 1976.

$P(C|S) \neq P(C|S^c)$